



Empowering Medical Diagnosis: A Machine Learning Approach for Symptom-Based Health Checker

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Abstract

AI-powered health checkers and apps for automated medical diagnosis have a lot of promise for a variety of applications. During pandemics, they can lessen the need for in-person patient-doctor interactions and offer vital medical advice in neglected rural areas. In this work, we demonstrate the creation of an expert system driven by machine learning on the web. This technology helps medical practitioners make better diagnostic decisions by supporting them and by offering accurate health forecasts and suggestions to the general population. Due to the lack of authentic medical datasets focusing on symptoms, we collected information from reputable medical sources. This enabled us to prioritize the medical diagnostic process, resulting in the compilation of a comprehensive list of illnesses and associated symptoms. This dataset played a key role in developing our health checker, which consisted of four primary parts: FrontEnd, Authentication module, BackEnd housing the machine learning module, and the Database. We constructed a dataset encompassing up to 415712 synthetic patients, 75 symptoms and risk factors, and 22 cough-related diagnoses. This dataset enabled the training and testing of supervised machine learning models to identify the most effective algorithm for implementation. The accuracy, performance and generalization ability of the utilized machine learning models were assessed using metrics including accuracy, F1-score and cross validation. Our work not only advances machine learning models but also addresses the pressing need for reliable medical datasets. The outcome of our efforts is a robust health checker, set to bring positive changes to diagnostic processes and healthcare accessibility as well as generalization and real-world applicability of our models. This highlights the critical role of dataset quality, especially with our ‘third dataset’ showcasing unparalleled performance across diverse medical scenarios with an accuracy superior to 99% and F1 score superior to 99% also for all the models. Stratified fivefold cross-validation also demonstrates positive results with an average accuracy and an average F1 score exceeding 99% for all models, thereby enhancing the reliability of our model evaluations and boosting confidence in the obtained metrics. In conclusion, our work propels the advancement of machine learning models, specifically addressing the imperative for reliable medical datasets. The result is a symptom-based health checker that demonstrates resilience, positioned to potentially contribute to advancements in diagnostics and improve accessibility to healthcare services.

Keywords Machine Learning · Artificial Intelligence · Medical Diagnosis · Symptom Checker · Dataset

1 Introduction

In recent years, the interest in using artificial intelligence (AI) in almost all fields of life was increased. This interest is even greater in the medical field and healthcare field where one of the examples of using AI is automatic medical diagnosis systems (AMDS) also known as health checkers or symptom checkers. AMDS, in general, are expert systems that can, among other objectives, determine whether a person has a particular condition based on a given set of symptoms, vital signs measures, laboratory test results, and X-ray findings, etc.

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The benefits of these systems are numerous. Firstly, in addition to improving healthcare decision-making, such systems aim essentially to lessen patient-doctor real-life first consultations and broaden access to medical care services [1, 2]. In fact, in 2013, the World Health Organization (WHO) has announced that the world is lacking 7.2 million medical workers and predicted that this number will reach 12.9 million in the next few years [3]. In 2019, WHO has also reported that many countries in the world have less than 1 doctor per 1000 people and called for the democratization of eHealth and telemedicine solutions [4]. Secondly, such solutions may be very useful and guiding tools for patients looking for a quick advice on medical issues and even help them detect early signs of a serious condition that may need immediate care. Indeed, numerous studies reveal that many people frequently use the internet to find quick health-related information for checking their health status. The majority of these people don't have sufficient health literacy that permits them to select the best information that fits their cases and consequently may be exposed to false or inaccurate information that can endanger their lives. Navigating on the web to check about health symptoms is very challenging for some people due to the use of highly technical jargon, the complex structure of many websites, and the lack of standards to publish online content [5–7].

For all these reasons, AMDS are indicated to help medical professionals triage patients and decide what management plan they should embark on and help public users in the process of self-diagnosis and guide them to the best and most adequate next step in treatment research [5, 8].

In the medical and biotechnology literature, many terms have been used to refer to an expert system such as “medical expert system” [9], “decision support and/or decision-making systems” [10], “diagnosis support system” [11], “symptom checker” [12], etc.

Generally, an expert System (ES) is an artificial intelligence-powered software that manipulates knowledge acquired from human experts in a non-procedural manner to solve problems within a specialized domain. It is basically composed of three components:

- Knowledge Base component: It holds the facts and the rules related to a particular field of knowledge.
- Inference Engine component: It receives input requests from the user via the user interface, matches the inputs with the rules stored in the knowledge base, selects and executes the right rule, and then, answers the user queries through the user interface.
- User Interface component: It permits the user to interact with the ES.

Ideally, an ES should have also an explanation facility module that can clarify the process of reasoning leading to

the proposed solution or the advanced advice [13]. Three main participants contribute in the building of an Expert System:

- Domain Expert (or specialized documented sources): A person who is specialized in a specific field of knowledge.
- Knowledge Engineer: A person whose mission is to collect knowledge from the domain experts and then translate this knowledge into a format that the inference engine understands. Many programming languages and tools have been developed to attend this goal like Lisp and Prolog, etc.
- End-User: A person who can be either an expert or not and who needs the ES solution for his queries. [13–15]

In the literature, many types of expert systems are advanced. These types are:

- Rule-based expert system: In this type of ES, knowledge is built of rules that are formulated as “if-then” statements. [16, 17]
- Fuzzy expert system: In contrast with the above-mentioned type that is based on the Boolean logic (“True” or “False”, “1” or “0”), this type is based on the fuzzy logic in which the truth values may be any number between “0” and “1”. In this case, the truth value represents the probability that a given statement is true or false [17].
- Frame-based expert system: Instead of using the Boolean logic or the fuzzy logic, in this type, knowledge is modeled into frames which are data structures that store knowledge and where every frame has its own name and group of attributes [17].
- Artificial neural network (ANN)/Machine Learning (ML) based expert system: It is a hybrid expert system that associates an ANN or another ML model and a rule-based expert system. It aims to benefit from the obvious advantages of both rule-based expert systems (explanation) and neural networks/machine learning (ability to learn, generalization, etc.). To build such a system, a dataset of real or simulated samples from which the models will be trained and tested is required. [17]
- Neuro-fuzzy expert system: It is a hybrid expert system that combines an ANN model and a fuzzy expert system. It aims to benefit from the ability of ANN to learn and deal with raw data and the ability of fuzzy systems to deal with reasoning using linguistic information gathered from domain experts. [17]
- Reinforcement learning powered expert system: Due to the limitations of the previously mentioned types of expert systems especially in the knowledge acquisition process, many researchers proposed to empower expert systems' inference component with reinforcement learn-

ing (RL) algorithms. RL permits an agent to learn the optimal behavior in an interactive given environment by trial and error. [18, 19]

Although AMDS's promise, a significant obstacle arises: there are not enough datasets available for symptoms-based diagnosis including illnesses involving the cough. Strong and varied datasets are necessary for the training and validation of these systems, which guarantees the correctness and dependability of AMDS development. Despite the burgeoning interest in AI-powered medical diagnostics, a notable hindrance is the scarcity of readily available datasets specifically tailored to symptoms, encompassing a wide array of illnesses. This dearth of comprehensive datasets poses a significant challenge to the advancement of AI applications in symptom-based medical diagnosis.

Our objective with the health checker is to provide a form of first aid in the diagnostic process based on symptoms. In other words, we don't have any secondary exam results (such as blood analysis results, which are generally real numbers, or radiographs, which are images) yet. The comparison between expert systems and machine learning shows that the machine learning-based expert system is a hybrid expert system that combines a machine learning model with a rule-based expert system. Our health checker is a hybrid expert system aiming to benefit from the obvious advantages of both rule-based expert systems (explanation) and machine learning (ability to learn, generalization, etc.). To build such a system, a dataset of real or simulated patients from which the models will be trained and tested is necessary. Machine learning models can also be continually improved and updated as new data becomes available, providing them with significant flexibility and scalability.

Within this framework, our study work entails the painstaking creation of a useful dataset, which is then finely entwined with artificial patient profiles that are produced from professional medical advice, clinical vignettes and suggestions given by active clinicians. Our steadfast commitment is directed at the creation of an intelligent online health assessment system that is smoothly powered by state-of-the-art machine learning algorithms, building upon the insights gained from this dataset. The ultimate goal is to easily include this advanced health checker into several healthcare ecosystems, such as medical office management systems, hospital and healthcare management systems, and telemedicine platforms. With its integrated system, which offers patients easily accessible and real-time health evaluations, medical diagnostics might undergo a revolution. Simultaneously, it aims to streamline the operations of healthcare professionals and institutions, ushering in a new era of efficiency in the healthcare domain. Furthermore, this health checker can be of great assistance in areas lacking qualified medical personnel. By providing a form of first

aid in the diagnostic process, it can help bridge the gap in healthcare access in such regions. It's designed to be user-friendly and does not require any specialized knowledge to use, making it a valuable tool for individuals in these areas to get efficient preliminary medical advice.

Recent innovations in healthcare robotics have indeed been impressive. However, these innovations primarily focus on assisting healthcare professionals in their tasks or improving sanitation. Our work, on the other hand, aims to empower both healthcare professionals and the general public by providing an accessible and accurate health checker. Another concern is the reliability and cost of existing robots. In fact, existing diagnostic robots are not 100% reliable and require constant updates and testing. Their high development costs limit their use in public health programs in disadvantaged countries and in educational settings. Our health checker aims to address these issues by providing a cost-effective, reliable and remote solution. However, it's important to note that while this tool can provide valuable insights, it's not a substitute for professional medical advice.

2 Methods

2.1 Medical Diagnostic Process

The work team is multidisciplinary, including a doctor who also holds a bachelor's degree in internet of things, and a senior researcher specialized in information and communication technologies. The doctor exploited her medical background to assure the accuracy of the collected medical knowledge. To understand the medical diagnostic process, we have consulted some medical books [20–22] and medically renowned websites like ViDAL website [23] and Mayo Clinic website [24]. We have also reviewed authentic clinical vignettes and engaged in discussions with several medical doctors in order to elucidate the real-life medical diagnosis process steps in general medicine. The results of this investigation are summarized in the corresponding results' section.

2.2 Dataset Exploration

In the early phases of our work, we have searched for an already existing public dataset that can meet our objective which is “medical diagnosis through patients' symptoms”. A dataset composed of 2 CSV files was found on Kaggle website [25]. The same dataset was found recently grouped in one csv file and published on Mendeley data repository [26]. The author mentioned that the dataset was gathered from reliable medical journals, clinical notes, and professional opinions following the approach of leading institutions specialized on public's health and preventive healthcare [26].

For analyzing this dataset, we have used Anaconda as code editor and we have been served with the pandas, numpy and sickit learn python libraries. To identify the most suitable model for implementation in our solution, we conducted comprehensive training and testing using the aforementioned dataset.

The dataset has been fed into various machine learning models to evaluate their performance. Using pandas and numpy, we were able to preprocess and manipulate the dataset to be prepared for training and testing. We trained the models on a subset of the dataset and then evaluated their performance using various evaluation metrics such as accuracy, recall, F1 score, etc. This allowed us to compare and to evaluate the models based on overall performance for our specific medical diagnosis task.

By conducting this comprehensive training and testing process, we ensured that the selected model was capable to efficiently use the patients' symptoms to make accurate medical diagnoses. This dataset exploration phase was crucial to guide us towards building a robust and accurate solution for medical diagnosis based on machine learning.

2.3 Selection and Evaluation of Machine Learning Models

In our study, we carefully considered the selection of machine learning algorithms with varied hyperparameters based on their suitability for symptom-based health checkers and medical diagnosis.

The specific machine learning algorithms were chosen for the study because they offer advantages such as accuracy, reliability, and generalization capabilities. The choice of model depends on the type of input data and the problem to be solved. In our case, it's a multi-classification problem dealing with binary data. It's important to note that the use of machine learning in this context allows us to handle this binary data effectively and identify patterns that may not be immediately apparent. The machine learning algorithms selected for our study have demonstrated performance in similar tasks in the existing literature. Firstly, we opted for supervised machine learning algorithms due to the availability of labeled training data. Supervised learning algorithms are well-suited for classification tasks, where the goal is to predict the presence or absence of a certain condition based on a set of symptoms. These algorithms learn from labeled examples and generalize to make predictions on unseen data. Secondly, we considered the complexity and non-linearity of medical data. Machine learning algorithms such as Support Vector Machines (SVM) and Random Forests (RF) have been widely used in medical diagnosis due to their ability to handle complex relationships and capture intricate patterns in the data. Here is an explanation of the choice for each selected model:

- Decision Trees model provide a rapid and effective method of categorizing data sets. They are reliable and represent gathered knowledge simply offering high classification accuracy. They have been used in different areas of medical decision making [27].
- Random Forest is an ensemble learning method that operates by constructing multiple decision trees. It provides better prediction accuracy compared to many other algorithms. It's particularly useful in medical diagnosis due to its ability to handle complex datasets with many features and instances such as rare disease diagnosis [28].
- The Multinomial Naive Bayes classification is a well-known supervised machine learning technique that is widely recognized for its simplicity and ease of implementation in classification tasks. One of its key advantages is its computational efficiency, making it suitable for large datasets and real-time applications. One of the reasons for its efficiency is that Naive Bayes relies on simple probability calculations. It can work well even with relatively small datasets because it makes the assumption of conditional independence between features. This assumption allows the model to simplify the calculations and make predictions based on straightforward probability estimations. Moreover, the model offers good interpretability thanks to its probabilistic nature. The model calculates the probabilities of different classes based on the observed features and uses Bayes' theorem to estimate the posterior probability. This allows users to understand why a particular classification decision was made and gain insights into the underlying probabilistic reasoning. The model offers a straightforward and efficient approach to classification tasks. Its simplicity, computational efficiency, and interpretable results make it a popular choice in various domains, particularly when dealing with real-time applications or different-size datasets [29].
- Logistic Regression is frequently used in medical research to classify, explain, or predict the values of some characteristic, behavior, or outcome. It's particularly useful when the variables are binary such as the case in our symptom datasets [30].
- K-Nearest Neighbors (KNN) is a powerful classification algorithm that has been used in various medical diagnosis tasks such as diagnosis of Parkinson and heart diseases [31, 32]. It's particularly effective when the dataset is not linearly separable [31, 32].
- Support Vector Classifier (SVC) is a powerful algorithm known for its effectiveness in high-dimensional spaces and its ability to handle complex relationships within the data. It is robust to overfitting, thanks to regularization techniques, and offers versatility through different kernel functions for modeling non-linear relationships. SVC performs well with small to medium-sized datasets,

and its strong theoretical foundation ensures reliable and robust predictions. Additionally, SVC can handle imbalanced data, making it suitable for various real-life scenarios [32].

The array of evaluated models includes:

- Mod1: DecisionTreeClassifier (),
- Mod2: RandomForestClassifier (max_depth = 2, random_state = 0),
- Mod3: RandomForestClassifier (max_depth = 3, random_state = 0),
- Mod4: RandomForestClassifier (max_depth = 4, random_state = 0),
- Mod5: RandomForestClassifier (max_depth = 5, random_state = 0),
- Mod6: RandomForestClassifier (max_depth = 6, random_state = 0),
- Mod7: RandomForestClassifier (max_depth = 7, random_state = 0),
- Mod8: RandomForestClassifier (max_depth = 10, random_state = 0),
- Mod9: MultinomialNB (),
- Mod10: LogisticRegression (),
- Mod11: KNeighborsClassifier (n_neighbors = 2),
- Mod12: KNeighborsClassifier (n_neighbors = 3),
- Mod13: KNeighborsClassifier (n_neighbors = 5),
- Mod14: KNeighborsClassifier (n_neighbors = 10),
- Mod15: SVC ().

To measure the performance of the models, we have been referred to the “Accuracy score” and the “F1 score”. The “Accuracy score” measures the proportion of the properly predicted instances among all of the instances. The

“F1 score” calculates the harmonic mean of the “Precision score” and the “Recall score”. The “Precision score” calculates the proportion of true positives among all the probable positive observations. It calculates the degree of how accurate is the output class’s prediction. The “Recall score” calculates the proportion of only true positives considering the total instances. In other words, it calculates the degree of the model’s ability to correctly identify a class. The results of the corresponding classification reports are summarized in Table 1 accompanied with a brief description of the found dataset. To obtain a more accurate estimation of the model prediction performances and to assess the models’ generalizability, we have been referred to the “stratified Kfold Cross Validation” (“stratified fivefold Cross Validation” in our case) which is a variety of KFold Cross Validation that maintain the same percentage of samples for each class as the one of the whole dataset. Using the stratified Kfold Cross Validation is very interesting especially in imbalanced datasets. The results of the “stratified fivefold Cross Validation” are summarized in Table 2 and Table 3.

2.4 Construction of Reliable New Datasets

After investigating the real-life medical diagnosis process in general medicine as mentioned above and taking into account the results of the ‘kaggle dataset’ analysis, we have been inspired by the principle of constructing a rule-based expert system (taking the knowledge from an expert and translating it into a format that a computer can understand) and the format of the data in the ‘kaggle dataset’, we proposed to construct datasets that are composed of diseases which are categorized according to the patient’s possible chief complaints and their possible symptoms based on the information taken from the consulted sources. On the whole,

Table 1 Classification reports of the used models for the ‘kaggle dataset’

Model		Accuracy (%)	F1_score (%)
Mod1	DecisionTreeClassifier ()	95.12	95.12
Mod2	RandomForestClassifier (max_depth = 2, random_state = 0)	85.36	83.54
Mod3	RandomForestClassifier (max_depth = 3, random_state = 0)	92.68	91.46
Mod4	RandomForestClassifier (max_depth = 4, random_state = 0)	95.12	94.31
Mod5	RandomForestClassifier (max_depth = 5, random_state = 0)	95.12	95.12
Mod6	RandomForestClassifier (max_depth = 6, random_state = 0)	95.12	95.12
Mod7	RandomForestClassifier (max_depth = 7, random_state = 0)	95.12	95.12
Mod8	RandomForestClassifier (max_depth = 10, random_state = 0)	95.12	95.12
Mod9	MultinomialNB ()	95.12	95.12
Mod10	LogisticRegression ()	95.12	95.12
Mod11	KNeighborsClassifier (n_neighbors = 2)	100	100
Mod12	KNeighborsClassifier (n_neighbors = 3)	100	100
Mod13	KNeighborsClassifier (n_neighbors = 5)	95.12	95.12
Mod14	KNeighborsClassifier (n_neighbors = 10)	95.12	95.12
Mod15	SVC ()	95.12	95.12

Table 2 Results of cross-validation accuracies of the used models in the “kaggle dataset”

Model		Stratified 5-Fold accuracy (%)					
		Split1	Split2	Split3	Split4	Split5	Mean Accuracy (Confidence Interval)
Mod1	DecisionTreeClassifier ()	99.49	99.49	99.49	99.49	99.59	100 (± 00)
Mod2	RandomForestClassifier (max_depth = 2, random_state = 0)	88.51	86.17	84.04	87.50	90.04	87 (± 04)
Mod3	RandomForestClassifier (max_depth = 3, random_state = 0)	89.73	89.32	87.50	94.81	97.35	92 (± 07)
Mod4	RandomForestClassifier (max_depth = 4, random_state = 0)	92.17	93.08	94.30	94.81	97.35	94 (± 04)
Mod5	RandomForestClassifier (max_depth = 5, random_state = 0)	96.74	97.25	96.95	97.25	97.35	97 (± 00)
Mod6	RandomForestClassifier (max_depth = 6, random_state = 0)	97.86	98.57	98.98	99.49	99.59	99 (± 01)
Mod7	RandomForestClassifier (max_depth = 7, random_state = 0)	98.88	99.08	99.08	99.49	99.59	99 (± 01)
Mod8	RandomForestClassifier (max_depth = 10, random_state = 0)	99.49	99.49	99.39	99.49	99.59	99 (± 00)
Mod9	MultinomialNB ()	99.49	99.49	99.49	99.49	99.59	100 (± 00)
Mod10	LogisticRegression ()	99.49	99.49	99.49	99.49	99.59	100 (± 00)
Mod11	KNeighborsClassifier (n_neighbors = 2)	97.66	96.95	95.83	95.02	95.12	96 (± 02)
Mod12	KNeighborsClassifier (n_neighbors = 3)	99.08	97.96	96.23	95.02	97.25	97 (± 03)
Mod13	KNeighborsClassifier (n_neighbors = 5)	99.49	99.49	99.49	99.49	99.59	100 (± 00)
Mod14	KNeighborsClassifier (n_neighbors = 10)	99.49	99.49	99.49	99.49	99.59	100 (± 00)
Mod15	SVC ()	99.49	99.49	99.49	99.49	99.59	100 (± 00)

we constructed three datasets (‘first dataset’, ‘second dataset’, and ‘third dataset’), and to address concerns about dataset limitations and overfitting, we have implemented several strategies:

- **Dataset expansion:** we have incrementally expanded our datasets by incorporating more symptoms and possible diseases to our first prototype dealing with conditions related to the cough chief complaint. This has allowed us to acquire new patient profiles.
- **High-quality data generation:** to enhance the reliability and accuracy of our datasets, we are using a combination of virtual patients and renowned medical sources. This approach ensures that our datasets are informed by credible medical knowledge and expert systems, resulting in high-quality data.
- **K-fold cross validation:** is a technique used to train and evaluate models. It splits the data into several parts and trains multiple algorithms on these parts to rely the model by excluding some data from the training process. It improves performance on new, unseen data and is particularly useful for limited datasets, as it allows effective training with a smaller amount of data. To obtain a more accurate estimate of model prediction

performance and assess its generalizability, we have used stratified K-fold cross-validation, which is particularly valuable when dealing with imbalanced datasets [33, 34].

The data in new constructed datasets is organized as follows:

- The columns contain the possible symptoms (features) and the possible diagnosis (target).
- The rows represent synthetic patients. Each row is a combination of some symptoms that may lead to a diagnosis.

In the first step of our study, we worked only on one chief complaint which is “Cough” and we used scientific and medical terms in English language for creating our content. The causes of “Cough” are numerous including Asthma, Bronchitis, Acute sinusitis, Chronic sinusitis, Pneumonia, Common cold, Flu, etc. [20–22] To accurately determine the most probable disease or diseases that may explain the user’s cough, our solution should inquire about the duration of the cough, the characteristics of the cough, some risk factors (age, medical history, allergies, etc.), and the associated symptoms.

Table 3 Cross-validation F1_scores results of the used models in the “kaggle dataset”

Model		Stratified 5-Fold F1_score (%)					
		Split1	Split2	Split3	Split4	Split5	Mean F1_score (Confidence Interval)
Mod1	DecisionTreeClassifier ()	99.24	99.24	99.24	99.24	99.39	99 (±00)
Mod2	RandomForestClassifier (max_depth=2, random_state=0)	87.22	84.09	81.86	84.71	87.09	85 (±04)
Mod3	RandomForestClassifier (max_depth=3, random_state=0)	87.25	87.59	85.09	93.07	96.44	90 (±08)
Mod4	RandomForestClassifier (max_depth=4, random_state=0)	90.64	91.77	93.05	93.07	96.44	93 (±04)
Mod5	RandomForestClassifier (max_depth=5, random_state=0)	95.76	97.01	96.07	96.29	96.44	96 (±01)
Mod6	RandomForestClassifier (max_depth=6, random_state=0)	97.59	98.30	98.74	99.24	99.39	99 (±01)
Mod7	RandomForestClassifier (max_depth=7, random_state=0)	98.62	98.84	98.84	99.24	99.39	99 (±01)
Mod8	RandomForestClassifier (max_depth=10, random_state=0)	99.24	99.24	99.14	99.24	99.39	99 (±00)
Mod9	MultinomialNB ()	99.24	99.24	99.24	99.24	99.39	99 (±00)
Mod10	LogisticRegression ()	99.24	99.24	99.24	99.24	99.39	99 (±00)
Mod11	KNeighborsClassifier (n_neighbors=2)	98.00	97.47	96.23	94.97	94.98	96 (±02)
Mod12	KNeighborsClassifier (n_neighbors=3)	99.02	98.31	96.75	94.97	97.10	97 (±03)
Mod13	KNeighborsClassifier (n_neighbors=5)	99.24	99.24	99.24	99.24	99.39	99 (±00)
Mod14	KNeighborsClassifier (n_neighbors=10)	99.24	99.24	99.24	99.24	99.39	99 (±00)
Mod15	SVC ()	99.24	99.24	99.24	99.24	99.39	99 (±00)

Taking this information into consideration and to quickly test our methodology, we have been able to construct, in a first phase, a ‘first dataset’ that is composed of 301 synthetic patients, 46 features describing the characteristics, the type and the associated symptoms of the cough, with 10 possible diagnoses as targets. Certainly, the causes of cough are more than 10 causes but as a first prototype, we have been contenting with this number. These 10 diagnoses are: ‘Influenza’, ‘Pneumonia’, ‘Asthma’, ‘Bronchitis’, ‘Common cold’, ‘Pertussis’, ‘Acute sinusitis’, ‘Gastroesophageal reflux disease’, ‘Chronic sinusitis’, ‘Bronchiolitis’. The distribution of samples in the ‘first dataset’ after the “train_test_split (Dataset, test_size=0.2, random_state=0)” is reported in Table 4. We used this dataset to train the same models trained with the ‘kaggle dataset’. The results of the different classification reports are summarized in Table 5 and the results of the “stratified fivefold Cross Validation” are summarized in Table 6 and Table 7.

In a second phase, we decided to reshape our ‘first dataset’ by adding some other features that are related essentially to the risk factors of diseases caused by the cough including age, gender, smoking, alcohol consumption, etc. This dataset becomes composed of 9572 rows presenting synthetic patients with 70 columns for the features and

Table 4 Distribution of the samples in the ‘first dataset’

Diagnosis	Number of samples in the trainset	Number of samples in the testset	Total
Influenza	3	1	4
Pneumonia	52	12	64
Asthma	20	4	24
Bronchitis	19	5	24
Common cold	5	3	8
Pertussis	6	6	12
Acute sinusitis	37	11	48
Gastroesophageal reflux disease	14	2	16
Chronic sinusitis	79	17	96
Bronchiolitis	5	0	5
Total	240	61	301

1 column for the target. The distribution of samples in the ‘second dataset’ after the “train_test_split (Dataset, test_size=0.2, random_state=0)” is reported in Table 8. We used this dataset to train the same models trained with the ‘kaggle dataset’ and the ‘first dataset’.

Table 5 Classification reports of the used models in the “first dataset”

Model		Accuracy (%)	F1_score (%)
Mod1	DecisionTreeClassifier ()	100	100
Mod2	RandomForestClassifier (max_depth=2, random_state=0)	78.68	72.71
Mod3	RandomForestClassifier (max_depth=3, random_state=0)	88.52	85.72
Mod4	RandomForestClassifier (max_depth=4, random_state=0)	98.36	98.23
Mod5	RandomForestClassifier (max_depth=5, random_state=0)	100	100
Mod6	RandomForestClassifier (max_depth=6, random_state=0)	100	100
Mod7	RandomForestClassifier (max_depth=7, random_state=0)	100	100
Mod8	RandomForestClassifier (max_depth=10, random_state=0)	100	100
Mod9	MultinomialNB ()	96.72	96.80
Mod10	LogisticRegression ()	100	100
Mod11	KNeighborsClassifier (n_neighbors=2)	96.72	96.80
Mod12	KNeighborsClassifier (n_neighbors=3)	100	100
Mod13	KNeighborsClassifier (n_neighbors=5)	93.44	93.11
Mod14	KNeighborsClassifier (n_neighbors=10)	90.16	89.90
Mod15	SVC ()	100	100

Table 6 Cross-validation accuracies results of the used models in the “first dataset”

Model		Stratified 5-Fold accuracy (%)					
		Split1	Split2	Split3	Split4	Split5	Mean Accuracy (Confidence Interval)
Mod1	DecisionTreeClassifier ()	100	100	100	100	100	100 (±00)
Mod2	RandomForestClassifier (max_depth=2, random_state=0)	89.58	91.66	85.41	91.66	89.58	90 (±05)
Mod3	RandomForestClassifier (max_depth=3, random_state=0)	95.83	95.83	93.75	95.83	91.66	95 (±03)
Mod4	RandomForestClassifier (max_depth=4, random_state=0)	95.83	95.83	100	100	93.75	97 (±05)
Mod5	RandomForestClassifier (max_depth=5, random_state=0)	100	100	100	100	95.83	99 (±03)
Mod6	RandomForestClassifier (max_depth=6, random_state=0)	100	100	100	100	95.83	99 (±03)
Mod7	RandomForestClassifier (max_depth=7, random_state=0)	100	100	100	100	95.83	99 (±03)
Mod8	RandomForestClassifier (max_depth=10, random_state=0)	100	100	100	100	95.83	99 (±03)
Mod9	MultinomialNB ()	100	100	100	100	95.83	99 (±03)
Mod10	LogisticRegression ()	100	100	100	100	95.83	99 (±03)
Mod11	KNeighborsClassifier (n_neighbors=2)	100	100	100	100	95.83	99 (±03)
Mod12	KNeighborsClassifier (n_neighbors=3)	100	100	93.75	97.91	95.83	97 (±05)
Mod13	KNeighborsClassifier (n_neighbors=5)	95.83	95.83	93.75	100	91.66	95 (±06)
Mod14	KNeighborsClassifier (n_neighbors=10)	93.75	91.66	91.66	97.91	89.58	93 (±06)
Mod15	SVC ()	100	100	100	100	95.83	99 (±03)

The results of the different classification reports are summarized in Table 9 and the results of the “stratified fivefold Cross Validation” are summarized in Table 10 and Table 11.

In a third phase, we carefully considered the risk of overfitting, especially when dealing with small datasets.

That’s why, we constructed and tested a ‘third dataset’ containing more conditions that can cause a cough. This ‘third dataset’ becomes composed of 415,712 rows presenting synthetic patients with 75 columns for the features and 1 column for the target. We used this dataset to

Table 7 Cross-validation F1_scores results of the used models in the “first dataset”

Model		Stratified 5-Fold F1_score (%)					
		Split1	Split2	Split3	Split4	Split5	Mean F1_score (Confidence Interval)
Mod1	DecisionTreeClassifier ()	100	100	100	100	100	100 (± 00)
Mod2	RandomForestClassifier (max_depth = 2, random_state = 0)	85.85	88.02	82.89	88.80	85.33	86 (± 05)
Mod3	RandomForestClassifier (max_depth = 3, random_state = 0)	93.91	93.91	91.79	93.91	87.99	92 (± 05)
Mod4	RandomForestClassifier (max_depth = 4, random_state = 0)	93.91	93.91	100	100	92.75	96 (± 06)
Mod5	RandomForestClassifier (max_depth = 5, random_state = 0)	100	100	100	100	94.79	99 (± 04)
Mod6	RandomForestClassifier (max_depth = 6, random_state = 0)	100	100	100	100	94.79	99 (± 04)
Mod7	RandomForestClassifier (max_depth = 7, random_state = 0)	100	100	100	100	94.79	99 (± 04)
Mod8	RandomForestClassifier (max_depth = 10, random_state = 0)	100	100	100	100	94.79	99 (± 04)
Mod9	MultinomialNB ()	100	100	100	100	94.79	99 (± 04)
Mod10	LogisticRegression ()	100	100	100	100	94.79	99 (± 04)
Mod11	KNeighborsClassifier (n_neighbors = 2)	100	100	100	100	94.79	99 (± 04)
Mod12	KNeighborsClassifier (n_neighbors = 3)	100	100	92.88	97.22	94.79	97 (± 06)
Mod13	KNeighborsClassifier (n_neighbors = 5)	94.83	94.84	93.29	100	90.67	95 (± 06)
Mod14	KNeighborsClassifier (n_neighbors = 10)	90.85	88.77	90.28	96.99	84.93	90 (± 08)
Mod15	SVC ()	100	100	100	100	94.79	99 (± 04)

Table 8 Distribution of the samples in the “second dataset”

Diagnosis	Number of samples in the trainset	Number of samples in the testset	Total
Influenza	111	25	136
Pneumonia	4612	1148	5760
Asthma	207	49	256
Bronchitis	348	84	432
Common cold	1026	254	1280
Pertussis	99	29	128
Acute sinusitis	308	76	384
Gastroesophageal reflux disease	113	27	140
Chronic sinusitis	811	213	1024
Bronchiolitis	22	10	32
Total	7657	1915	9572

train the same models trained with the ‘kaggle dataset’, the ‘first dataset’ and the ‘second dataset’. The distribution of samples in the ‘second dataset’ after the ‘train_test_split (Dataset, test_size = 0.2, random_state = 0)’ is reported in Table 12. The results of the different classification reports are summarized in Table 13 and the results of the stratified

fivefold cross validation are summarized in Table 14 and Table 15.

Figure 2 and Fig. 3 show the “accuracy” and the “F1_score” performance variations throughout the different used models in the four datasets. The Figs. 4–11 highlight the results stemming from the stratified fivefold cross-validation for the four datasets.

2.5 Solution Architecture

The architecture of our solution is composed of four main components: FrontEnd, Authentication module, BackEnd and the Database. The Front-End represents the user interface that permits the user to access our solution’s functionalities including the interaction with the expert system engine. The user requests are received at this level and redirected to the appropriate endpoints in the server side.

The Back-End provides the Front-End with all the necessary endpoints to perform the required tasks. The ML module, part of the Back-End, is composed of the inference engine and the knowledge base. The knowledge base holds the expert domain knowledge. In our work, the knowledge base is conceived to fit a machine learning-based expert system, so that it is composed of a dataset of patients with

Table 9 Classification reports of the used models in the “second dataset”

Model		Accuracy (%)	F1_score (%)
Mod1	DecisionTreeClassifier ()	100	100
Mod2	RandomForestClassifier (max_depth=2, random_state=0)	88.72	84.08
Mod3	RandomForestClassifier (max_depth=3, random_state=0)	97.91	97.53
Mod4	RandomForestClassifier (max_depth=4, random_state=0)	100	100
Mod5	RandomForestClassifier (max_depth=5, random_state=0)	100	100
Mod6	RandomForestClassifier (max_depth=6, random_state=0)	100	100
Mod7	RandomForestClassifier (max_depth=7, random_state=0)	100	100
Mod8	RandomForestClassifier (max_depth=10, random_state=0)	100	100
Mod9	MultinomialNB ()	100	100
Mod10	LogisticRegression ()	100	100
Mod11	KNeighborsClassifier (n_neighbors=2)	100	100
Mod12	KNeighborsClassifier (n_neighbors=3)	100	100
Mod13	KNeighborsClassifier (n_neighbors=5)	100	100
Mod14	KNeighborsClassifier (n_neighbors=10)	100	100
Mod15	SVC ()	100	100

Table 10 Cross-validation accuracies results of the used models in the “second dataset”

Model		Stratified 5-Fold accuracy (%)					
		Split1	Split2	Split3	Split4	Split5	Mean Accuracy (Confidence Interval)
Mod1	DecisionTreeClassifier ()	100	100	100	100	100	100 (±00)
Mod2	RandomForestClassifier (max_depth=2, random_state=0)	88.77	88.77	88.76	88.76	88.76	89 (±00)
Mod3	RandomForestClassifier (max_depth=3, random_state=0)	98.36	97.84	98.04	98.30	98.56	98 (±01)
Mod4	RandomForestClassifier (max_depth=4, random_state=0)	100	100	100	100	100	100 (±00)
Mod5	RandomForestClassifier (max_depth=5, random_state=0)	100	100	100	100	100	100 (±00)
Mod6	RandomForestClassifier (max_depth=6, random_state=0)	100	100	100	100	100	100 (±00)
Mod7	RandomForestClassifier (max_depth=7, random_state=0)	100	100	100	100	100	100 (±00)
Mod8	RandomForestClassifier (max_depth=10, random_state=0)	100	100	100	100	100	100 (±00)
Mod9	MultinomialNB ()	100	100	100	100	100	100 (±00)
Mod10	LogisticRegression ()	100	100	100	100	100	100 (±00)
Mod11	KNeighborsClassifier (n_neighbors=2)	100	100	100	100	100	100 (±00)
Mod12	KNeighborsClassifier (n_neighbors=3)	100	100	100	100	100	100 (±00)
Mod13	KNeighborsClassifier (n_neighbors=5)	100	100	100	100	100	100 (±00)
Mod14	KNeighborsClassifier (n_neighbors=10)	100	100	100	100	100	100 (±00)
Mod15	SVC ()	100	100	100	100	100	100 (±00)

their experienced symptoms and the related diagnosis and it is saved in XLSX file. As in the medical diagnosis process, each chief complaint is associated with some specified questions and to help the inference engine proceed with the allocated task as rapidly as possible, we proposed to divide

the global knowledge base into as many knowledge bases as the possible number of chief complaints as illustrated in Fig. 1. The inference engine holds the reasoning process leading to the predictions of the system. It necessitates the knowledge stored in the knowledge base to produce and then

Table 11 Cross-validation F1_scores results of the used models in the “second dataset”

Model		Stratified 5-Fold F1_score (%)					
		Split1	Split2	Split3	Split4	Split5	Mean F1_score (Confidence Interval)
Mod1	DecisionTreeClassifier ()	100	100	100	100	100	100 (± 00)
Mod2	RandomForestClassifier (max_depth = 2, random_state = 0)	84.18	84.18	84.18	84.18	84.18	84 (± 00)
Mod3	RandomForestClassifier (max_depth = 3, random_state = 0)	98.13	97.52	97.79	98.09	98.37	98 (± 01)
Mod4	RandomForestClassifier (max_depth = 4, random_state = 0)	100	100	100	100	100	100 (± 00)
Mod5	RandomForestClassifier (max_depth = 5, random_state = 0)	100	100	100	100	100	100 (± 00)
Mod6	RandomForestClassifier (max_depth = 6, random_state = 0)	100	100	100	100	100	100 (± 00)
Mod7	RandomForestClassifier (max_depth = 7, random_state = 0)	100	100	100	100	100	100 (± 00)
Mod8	RandomForestClassifier (max_depth = 10, random_state = 0)	100	100	100	100	100	100 (± 00)
Mod9	MultinomialNB ()	100	100	100	100	100	100 (± 00)
Mod10	LogisticRegression ()	100	100	100	100	100	100 (± 00)
Mod11	KNeighborsClassifier (n_neighbors = 2)	100	100	100	100	100	100 (± 00)
Mod12	KNeighborsClassifier (n_neighbors = 3)	100	100	100	100	100	100 (± 00)
Mod13	KNeighborsClassifier (n_neighbors = 5)	100	100	100	100	100	100 (± 00)
Mod14	KNeighborsClassifier (n_neighbors = 10)	100	100	100	100	100	100 (± 00)
Mod15	SVC ()	100	100	100	100	100	100 (± 00)

translate the rules that form the system’s prediction and that are represented in the shape of a truth table in our case.

The database is used to store users' information and facilitate signing up and signing in processes. The results of the system’s predictions or the doctors’ comments are also stored in the database to be fetched after that upon request. For the Authentication module, the session-based authentication method is adopted.

2.6 Clinical Cases Testing

After implementing our web-based solution using the “third dataset”, we proceeded to test some reported clinical cases from the literature in order to quickly check its possible applicability in real life. This does not dispense the necessity of a serious and rigorous clinical trial on a representative sample of real patients to ensure the safety and effectiveness of our symptom checker which can be the subject of a further research. We selected five clinical cases which are related to the cough chief complaint and for which the diagnosis is explicitly advanced. The diagnoses that have been tested are: Chronic Obstructive Pulmonary Disease COPD (Case 1) [35], Heart Failure (Case 2) [36], Pulmonary Embolism

(Case 3) [37], Cystic Fibrosis (Case4) [38] and Lung Cancer (Case 5) [39]. The results are highlighted in Table 16.

3 Results

3.1 Medical Diagnostic Process

Medical diagnostic process is the clinical reasoning that aims to find out the disease that can explain a person's “symptoms and signs”. In order to make the right medical diagnosis, doctors should follow a series of logical steps that are detailed below [20]:

- Step 1: Data acquisition

In the initial interaction with the patient, the physician (doctor) begins with interviewing the patient about his chief complaint and his history that includes age, past illnesses, current symptoms, demographics, current drugs, etc. Among the possible chief complaints that a patient can suffer from, we can cite: Cough, Headache, Chest Pain, Shortness of Breath, Dizziness, Fatigue, Fever, Arm Pain, Loss of Memory, Red Eye, etc. [20–22]

Table 12 Distribution of the samples in the ‘third dataset’

Diagnosis	Number of samples in the trainset	Number of samples in the testset	Total
Influenza	157354	39254	196608
Heart failure	39392	9760	49152
Covid19	39155	9997	49152
Allergic rhinitis	19895	4937	24832
Pulmonary embolism	19756	4820	24576
Common cold	19588	4988	24576
COPD	9902	2386	12288
Croup	4905	1239	6144
Sarcoidosis	3682	926	4608
Chronic sinusitis	3204	892	4096
Cystic fibrosis	2503	569	3072
Emphysema	2437	635	3072
Chronic bronchitis	2411	661	3072
Gastroesophageal reflux disease	1854	450	2304
Laryngitis	1232	304	1536
Pneumonia	1230	306	1536
Acute bronchitis	1217	319	1536
Acute sinusitis	805	219	1024
Lung cancer	794	166	960
Pertussis	461	115	576
Bronchiolitis	396	116	512
Asthma	396	84	480
Total	332569	83143	415712

- Step 2 and Step 3: Accurate problem representation and differential diagnosis development

Focusing on the mentioned chief complaint, and based on the patient’s history and the identified pivotal points during

this interview, physicians do not only focus on determining the patient's illness but they organize and prioritize the patient's questions into a discovery-confirmation framework in order to make a differential diagnosis (a reasonable short list of diseases to be considered). [20]

- Step 4: Prioritizing the differential diagnosis

At any time during the interaction, the physician must develop a differential diagnosis that can be modified depending on the patient's response. In fact, not all diagnoses in a given differential diagnosis are similarly important. The physician should select:

- “Leading hypothesis” (Disease that has the highest probability)
- “Must not miss hypothesis” (Disease with lower probability than that of the leading hypothesis but likely enough to be actively sought in the patient)
- “Active alternative hypotheses” (Not excluded diseases which are not likely enough to be sought and tested immediately)
- “Excluded hypotheses” (Excluded diseases from the differential list) [20]

- Step 5: Testing the above-mentioned hypotheses

Next, at the usual face-to-face clinical meeting, a physical examination will be performed and some required diagnostic tests will be ordered to test the hypotheses and confirm the diagnosis. [20]

- Step 6 and Step 7: Reviewing and reprioritizing the differential diagnosis

Table 13 Classification reports of the used models in the ‘third dataset’

Model		Accuracy (%)	F1_score (%)
Mod1	DecisionTreeClassifier ()	99.97	99.97
Mod2	RandomForestClassifier (max_depth=2, random_state=0)	77.55	70.68
Mod3	RandomForestClassifier (max_depth=3, random_state=0)	85.88	81.06
Mod4	RandomForestClassifier (max_depth=4, random_state=0)	90.60	87.63
Mod5	RandomForestClassifier (max_depth=5, random_state=0)	95.86	94.78
Mod6	RandomForestClassifier (max_depth=6, random_state=0)	97.72	97.11
Mod7	RandomForestClassifier (max_depth=7, random_state=0)	98.66	98.27
Mod8	RandomForestClassifier (max_depth=10, random_state=0)	99.61	99.54
Mod9	MultinomialNB ()	99.15	99.14
Mod10	LogisticRegression ()	99.98	99.98
Mod11	KNeighborsClassifier (n_neighbors=2)	99.92	99.92
Mod12	KNeighborsClassifier (n_neighbors=3)	99.96	99.96
Mod13	KNeighborsClassifier (n_neighbors=5)	99.96	99.96
Mod14	KNeighborsClassifier (n_neighbors=10)	99.96	99.96
Mod15	SVC ()	99.98	99.98

Table 14 Results of cross validation accuracies of the used models in the ‘third dataset’

Model		Stratified 5-Fold accuracy (%)					
		Split1	Split2	Split3	Split4	Split5	Mean Accuracy (Confidence Interval)
Mod1	DecisionTreeClassifier ()	99.96	99.96	99.96	99.96	99.96	100 (±00)
Mod2	RandomForestClassifier (max_depth=2, random_state=0)	77.82	77.71	77.80	77.78	77.79	78 (±00)
Mod3	RandomForestClassifier (max_depth=3, random_state=0)	85.55	85.62	85.95	85.85	86.03	86 (±00)
Mod4	RandomForestClassifier (max_depth=4, random_state=0)	90.92	90.85	90.94	91.00	90.86	91 (±00)
Mod5	RandomForestClassifier (max_depth=5, random_state=0)	96.37	96.26	96.22	96.20	96.19	96 (±00)
Mod6	RandomForestClassifier (max_depth=6, random_state=0)	97.75	97.78	97.73	97.75	97.83	98 (±00)
Mod7	RandomForestClassifier (max_depth=7, random_state=0)	98.66	98.76	98.72	98.71	98.66	99 (±00)
Mod8	RandomForestClassifier (max_depth=10, random_state=0)	99.60	99.64	99.63	99.61	99.63	100 (±00)
Mod9	MultinomialNB ()	99.18	99.12	99.16	99.16	99.14	99 (±00)
Mod10	LogisticRegression ()	99.98	99.98	99.98	99.97	99.95	100 (±00)
Mod11	KNeighborsClassifier (n_neighbors=2)	99.92	99.91	99.90	99.91	99.90	100 (±00)
Mod12	KNeighborsClassifier (n_neighbors=3)	99.95	99.96	99.97	99.96	99.94	100 (±00)
Mod13	KNeighborsClassifier (n_neighbors=5)	99.95	99.95	99.97	99.95	99.94	100 (±00)
Mod14	KNeighborsClassifier (n_neighbors=10)	99.95	99.94	99.96	99.96	99.94	100 (±00)
Mod15	SVC ()	99.97	99.98	99.98	99.98	99.96	100 (±00)

Then, if the final diagnosis is not reached yet, the differential diagnosis must be reviewed and reprioritized many times until the causes of the patient’s symptoms are detected and the final diagnosis can be confirmed [20].

Our health checker provides a differential diagnosis, offering a list of potential ranked diseases. It’s like a friendly guide, acknowledging the complexity of symptoms and providing a range of possibilities, not just a single diagnosis. This tool is particularly useful in areas with limited healthcare professionals, offering initial guidance based on symptoms. It’s like a virtual doctor, helping to identify potential health issues. However, it’s designed to support, not replace, professional medical advice, providing a preliminary diagnosis that needs validation. Our health checker is also a handy tool for doctors, offering a preliminary diagnosis to help them focus on the right tests and treatments.

3.2 Dataset

3.2.1 ‘kaggle dataset’

The ‘kaggle dataset’ is composed of 2 CSV files [25]. One of these files is for training ML models and the other one is for testing ML models. Each file has 133 columns divided

as follows: 132 columns representing different symptoms that a person can experience and 01 column representing medical diagnosis. A value of “0” or “1” was allocated to each symptom based on whether this symptom is associated with the diagnosed disease or not. In total, 42 diseases can be diagnosed. The training dataset is composed of 4920 rows (samples) and the testing dataset is composed of 41 rows (samples). The owner of this dataset is a researcher at Nirma University, Ahmedabad, Gujarat, India [40]. No other information about the origin of this dataset can be gathered. When observing the classification reports (Table 1), we noticed that all models exhibit an accuracy score of 95.12% and an F1_score of 95.12% also after adjusting the required hyperparameters. Additionally, we observed that both the average accuracy (Table 2) and the average F1_score (Table 3) of the stratified fivefold cross-validation are notably above 95% for all models after adjusting the required hyperparameters.

3.2.2 ‘first dataset’

Table 4 reports the distribution of samples in the ‘first dataset’. Table 5 summarizes the results of the different classification reports of the ‘first dataset’. Table 6 reports the

Table 15 Results of cross-validation F1_scores of the used models in the ‘‘third dataset’’

Model		Stratified 5-Fold F1_score (%)					
		Split1	Split2	Split3	Split4	Split5	Mean F1_score (Confidence Interval)
Mod1	DecisionTreeClassifier ()	99.96	99.96	99.96	99.96	99.96	100 (± 00)
Mod2	RandomForestClassifier (max_depth = 2, random_state = 0)	70.99	70.86	70.99	70.93	70.98	71 (± 00)
Mod3	RandomForestClassifier (max_depth = 3, random_state = 0)	80.73	80.76	81.17	81.06	81.24	81 (± 00)
Mod4	RandomForestClassifier (max_depth = 4, random_state = 0)	88.10	87.98	88.04	88.25	88.08	88 (± 00)
Mod5	RandomForestClassifier (max_depth = 5, random_state = 0)	95.47	95.36	95.27	95.24	95.25	95 (± 00)
Mod6	RandomForestClassifier (max_depth = 6, random_state = 0)	97.13	97.15	97.10	97.14	97.23	97 (± 00)
Mod7	RandomForestClassifier (max_depth = 7, random_state = 0)	98.30	98.39	98.36	98.32	98.30	98 (± 00)
Mod8	RandomForestClassifier (max_depth = 10, random_state = 0)	99.54	99.59	99.59	99.55	99.58	100 (± 00)
Mod9	MultinomialNB ()	99.16	99.11	99.14	99.14	99.13	99 (± 00)
Mod10	LogisticRegression ()	99.98	99.98	99.98	99.97	99.95	100 (± 00)
Mod11	KNeighborsClassifier (n_neighbors = 2)	99.92	99.91	99.90	99.91	99.91	100 (± 00)
Mod12	KNeighborsClassifier (n_neighbors = 3)	99.95	99.96	99.97	99.96	99.94	100 (± 00)
Mod13	KNeighborsClassifier (n_neighbors = 5)	99.95	99.95	99.97	99.95	99.94	100 (± 00)
Mod14	KNeighborsClassifier (n_neighbors = 10)	99.95	99.94	99.96	99.96	99.94	100 (± 00)
Mod15	SVC ()	99.97	99.98	99.98	99.98	99.96	100 (± 00)

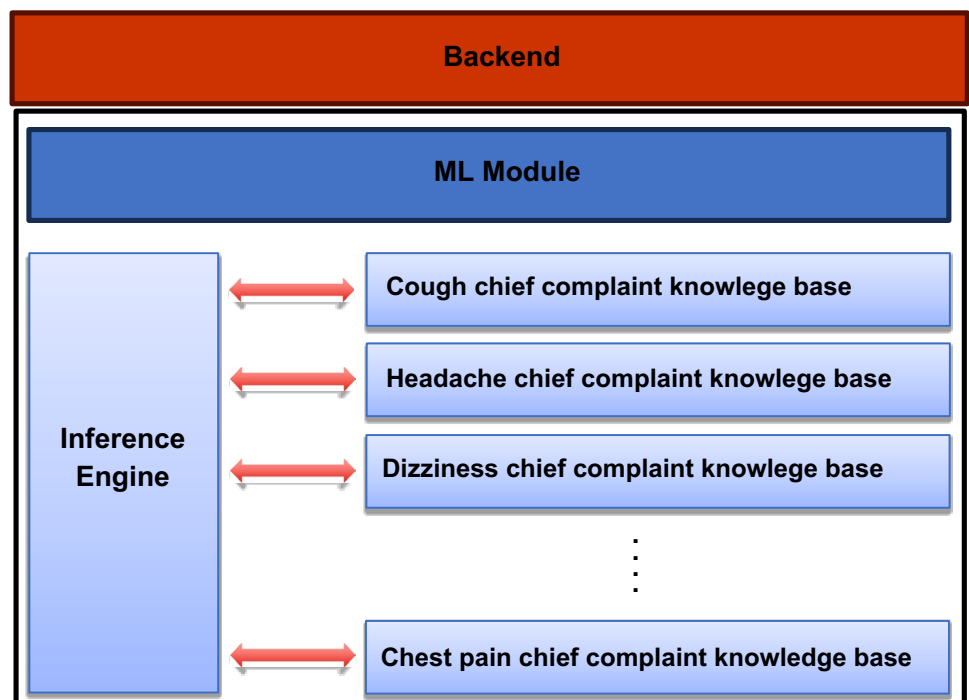
Fig. 1 Architecture of the machine learning module

Table 16 Results of clinical cases testing

		Clinical cases				
		Case 1	Case 2	Case 3	Case 4	Case 5
Mod1	PRS	no	no	no	no	yes
	Rank	*	*	*	*	1
Mod2	PRS	yes	yes	yes	yes	yes
	Rank	7	2	4	11	14
Mod3	PRS	yes	yes	yes	yes	yes
	Rank	7	2	3	11	9
Mod4	PRS	yes	yes	yes	yes	yes
	Rank	5	2	2	11	8
Mod5	PRS	yes	yes	yes	yes	Yes
	Rank	2	2	1	10	4
Mod6	PRS	yes	yes	yes	yes	yes
	Rank	1	4	1	10	3
Mod7	PRS	yes	yes	yes	yes	yes
	Rank	1	7	1	10	3
Mod8	PRS	yes	yes	yes	yes	yes
	Rank	1	5	1	9	2
Mod9	PRS	yes	yes	yes	yes	yes
	Rank	1	11	2	4	1
Mod10	PRS	yes	yes	yes	yes	yes
	Rank	1	10	2	11	1
Mod11	PRS	yes	no	yes	no	no
	Rank	1	*	1	*	*
Mod12	PRS	yes	no	yes	no	no
	Rank	1	*	1	*	*
Mod13	PRS	yes	no	yes	no	no
	Rank	1	*	2	*	*
Mod14	PRS	yes	no	yes	no	yes
	Rank	1	*	1	*	3
Mod15	PRS	yes	yes	yes	yes	yes
	Rank	2	10	2	11	1

“Accuracy scores” and the mean accuracy of the different splits stemming from the stratified 5-Fold cross validation for the ‘first dataset’.

Table 7 reports the “F1 scores” and the mean F1_score of the different splits stemming from the stratified 5-Fold cross validation for the ‘first dataset’. When observing the classification reports, we noticed that all of the models may show an accuracy score equal to 100% as well as an F1_score equal to 100% after adjusting the required hyperparameters such as in the case of the RandomForestClassifier() and except for the MultinomialNB() showing an accuracy score equal to 96.72% and an F1_score equal to 96.8%. We noticed also that the average accuracy of the stratified fivefold cross-validation is higher than 95% for all the models (Table 6) and the average F1_score of the stratified fivefold cross-validation is higher than 95% for all the models also after adjusting the required hyperparameters (Table 7).

3.2.3 ‘second dataset’

Table 8 reports the distribution of samples in the ‘second dataset’. The results of the different classification reports of the ‘second dataset’ are summarized in Table 9.

Table 10 reports the “Accuracy scores” and the mean accuracy of the different splits stemming from the stratified k-fold cross validation for the ‘second dataset’. Table 11 reports the “F1 scores” and the mean F1_score of the different splits stemming from the stratified 5-Fold cross validation for the ‘second dataset’.

As revealed in the classification reports (Table 9), all of the models may show an accuracy score equal to 100% as well as an F1_score equal to 100% after adjusting the required hyperparameters for the ‘second dataset’. Additionally, we noticed that both the accuracy score and F1 score have improved, and both the average accuracy and the

average F1_score of the stratified fivefold cross-validation (Table 10 and Table 11) are 100% for all models, except for RandomForestClassifier (max_depth=2) and RandomForestClassifier (max_depth=3), which have accuracy scores of 89% and 98% and F1_scores of 84% and 98%, respectively.

3.2.4 ‘third dataset’

Table 12 reports the distribution of samples in the ‘third dataset’. Table 13 summarizes the results of the different classification reports of the ‘third dataset’. Table 14 reports the “Accuracy scores” and the mean accuracy of the different splits stemming from the stratified 5-Fold cross validation for the ‘third dataset’. Table 15 reports the “F1 scores” and the mean F1_score of the different splits stemming from the stratified 5-Fold cross validation for the ‘second dataset’.

When observing the latter classification reports (Table 13), we noticed that, when using the ‘third dataset’, all of the models demonstrate an accuracy score exceeding 99% after adjusting the required hyperparameters, and an F1 score surpassing 99% after adjusting the required hyperparameters, except for RandomForestClassifier with some low tree depths. Equally, we noticed that both the average accuracy and the average F1_score of the stratified fivefold cross-validation (Table 14 and Table 15) are 100% for all models except for the MultinomialNB() showing an average accuracy score equal to 99% and an average F1_score equal to 99%.

When observing the latter classification reports (Fig. 2 and Fig. 3), we noticed that, in the case of using the ‘second dataset’, all of the models may show an accuracy score equal

to 100% as well as an F1_score equal to 100% after adjusting the required hyperparameters. We noticed also, that both Accuracy score and F1_score seem be ameliorated. When evaluating the performance of the different machine learning models (Fig. 2 and Fig. 3) across the four datasets (‘kaggle dataset’, dataset1, dataset2, and dataset3), we observe slightly varying results.

For ‘kaggle dataset’ and dataset1, all models achieved excellent accuracy and F1 score, indicating perfect performance due to the structured data and the relatively small datasets.

Regarding dataset2, performances vary slightly across models. RandomForestClassifier models with varying tree depths (up to 7), as well as LogisticRegression, MultinomialNB, and SVC, all achieved perfect accuracy and F1 scores. However, the RandomForestClassifier with a tree depth of 2 showed slightly lower performance with a precision of 88.72% and an F1 score of 84.08%.

For dataset3, the overall trend shows a progressive improvement in performance as the tree depth in the RandomForestClassifier increases. The lowest scores are observed with the RandomForestClassifier with a tree depth of 2, while the highest scores are achieved with a tree depth of 10, where all models reached an excellent accuracy and F1 score. Generally, the models performed well across the four datasets with some variations based on data size and model hyperparameters.

Furthermore, the results were rather stable and reliable. As cross-validation using stratified fivefold was run on the four datasets (Fig. 4, Fig. 5, Fig. 6, Fig. 7, Fig. 8, Fig. 9, Fig. 10 and Fig. 11) it turned out that the results are rather uniform. For ‘kaggle dataset’, cross validation

Fig. 2 Accuracy performance

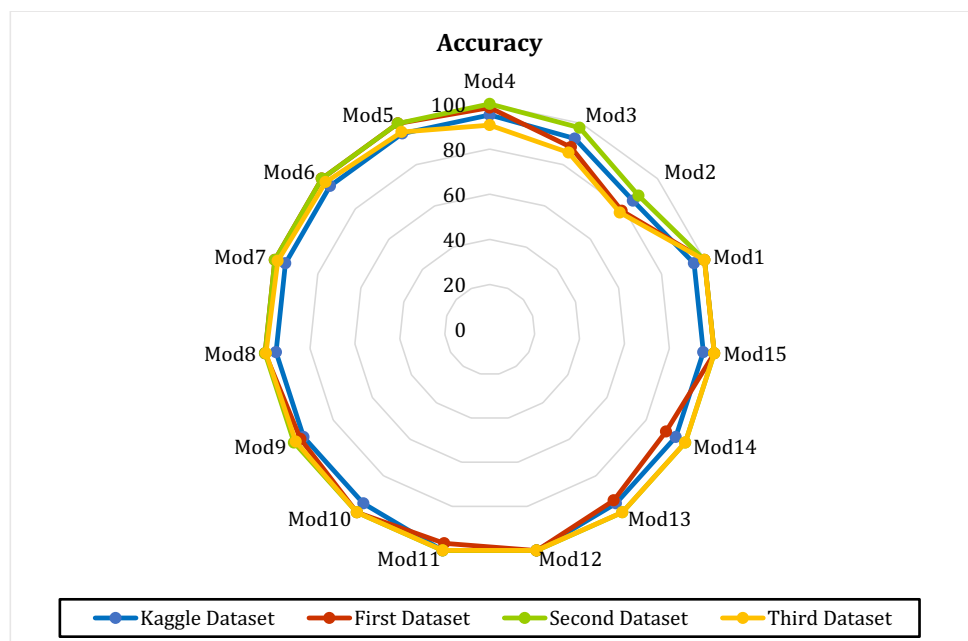
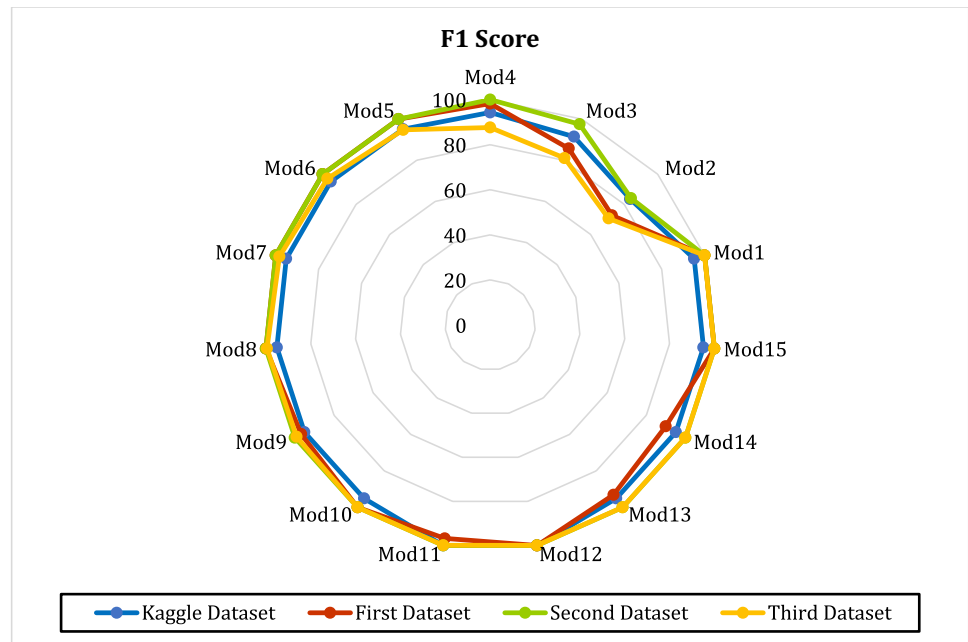


Fig. 3 F1_score performance

on 5 splits for all models showed high accuracy and high F1 score, from about 84% to 100% for both metrics. The model with a higher tree depth for RandomForestClassifier was the one that has a better score, which should mean a better ability to catch the complexity in the data. The results were rather robust for ‘first dataset’. The level of accuracy was about 100% on most of the splits. Once again, the scores were the best for the RandomForestClassifier with higher depths of the tree. In the case of the ‘second dataset’ and ‘third dataset’, all the models performed perfectly on all the cross-validation splits.

Finally, for ‘third dataset’, most models’ cross-validation scores were also high; however, for low tree depths in RandomForestClassifiers, the performance was slightly lower, hence the models may face difficulty generalizing the complexities of the data. In conclusion, fivefold cross-validation effectively evaluated the models’ performance in various datasets and provided a relative measure of how well the models generalize and avoid overfitting on training data. Thus, using stratified k-fold cross-validation strengthens the reliability of our model evaluations and boosts confidence in the obtained metrics.

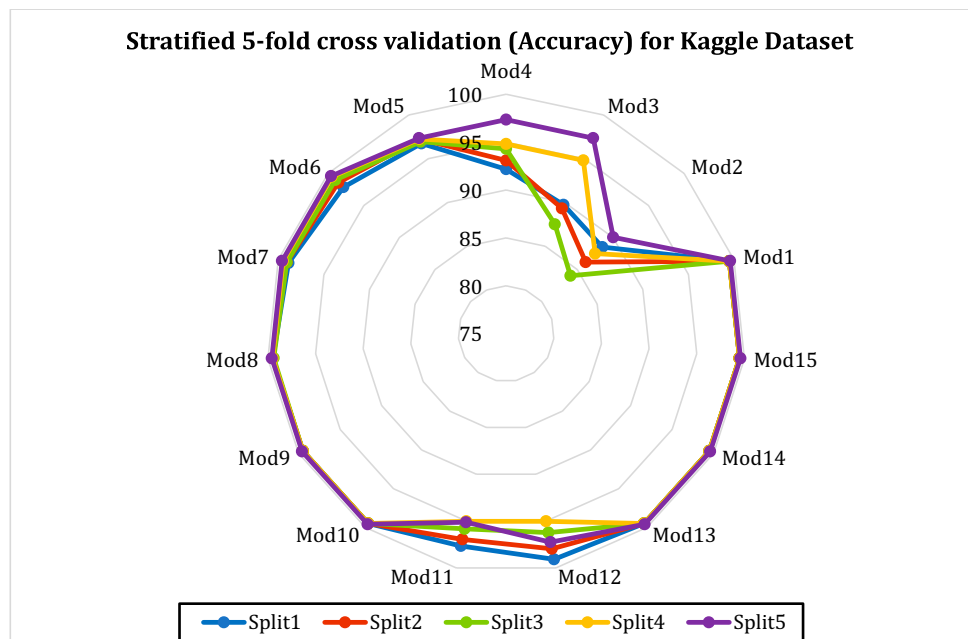
Fig. 4 Stratified 5-Fold Cross Validation (Accuracy) for ‘kaggle dataset’

Fig. 5 Stratified 5-Fold Cross Validation (Accuracy) for ‘first dataset’

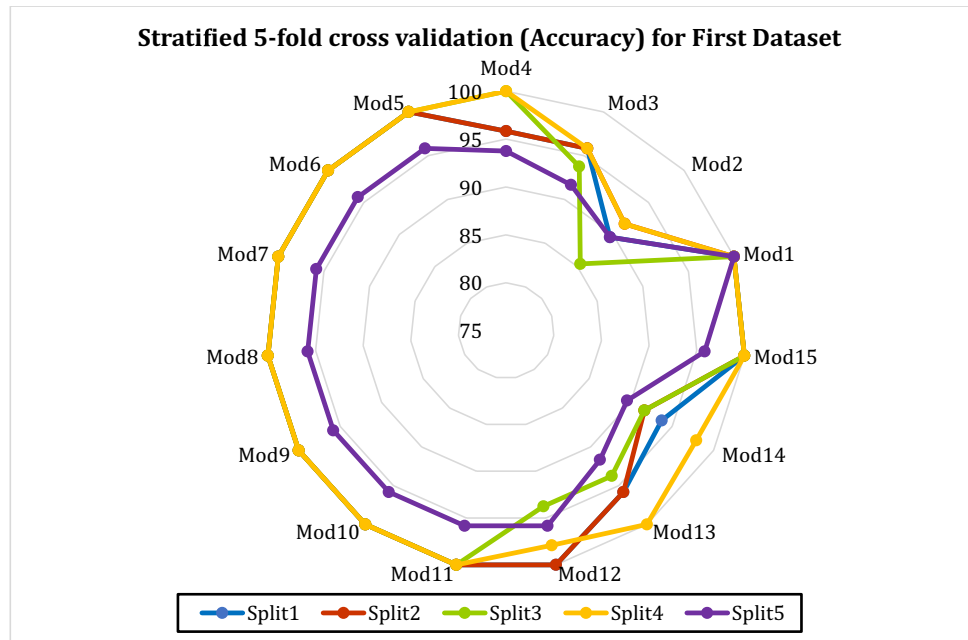
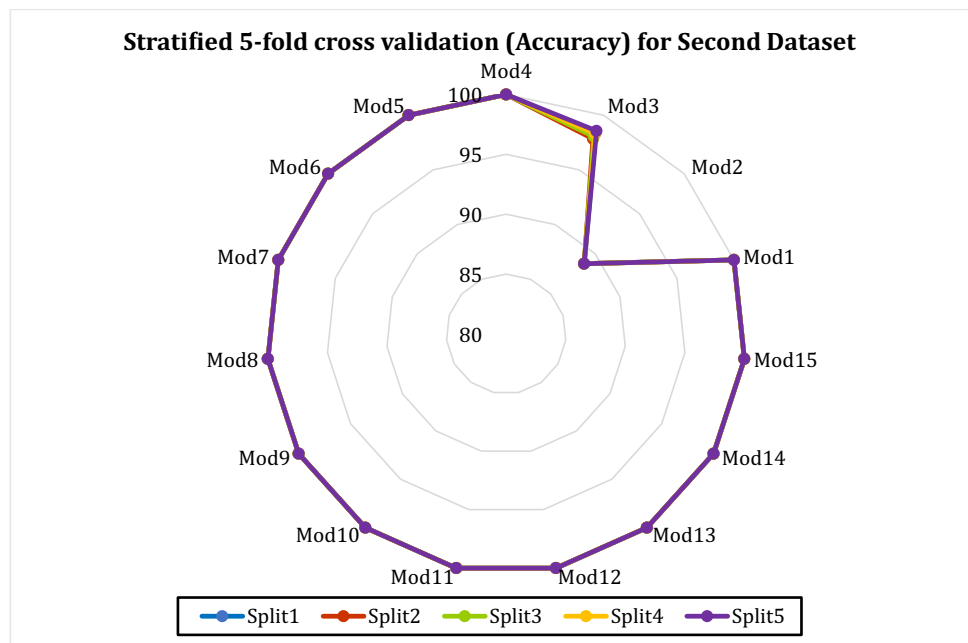


Fig. 6 Stratified 5-Fold Cross Validation (Accuracy) for ‘second dataset’



3.3 Solution Architecture

Figure 12 shows the home page of our solution. The user must press the “Get access” button to be redirected to the login page.

Figure 13 shows the login interface. The user must enter his email and password to access his corresponding user interfaces.

Figure 14 illustrates the patient’s user interface showing the “Consultation of the Health Checker” functionality. The Patient must select a chief complaint and then, he must click on “Continue” button to be redirected to the selected chief complaint’s form. Figure 15 and Fig. 16 show the “Cough chief complaint” form. The Patient must select the items that fit his case and then, he must click on “Continue” button. Figure 17 shows the Health Checker

Fig. 7 Stratified 5-Fold Cross Validation (Accuracy) for ‘third dataset’

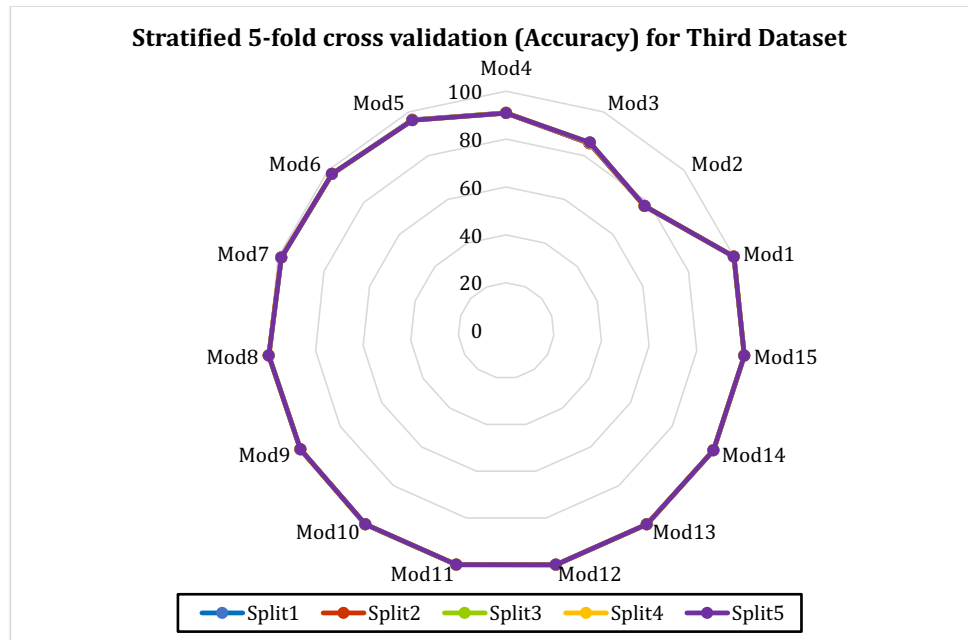
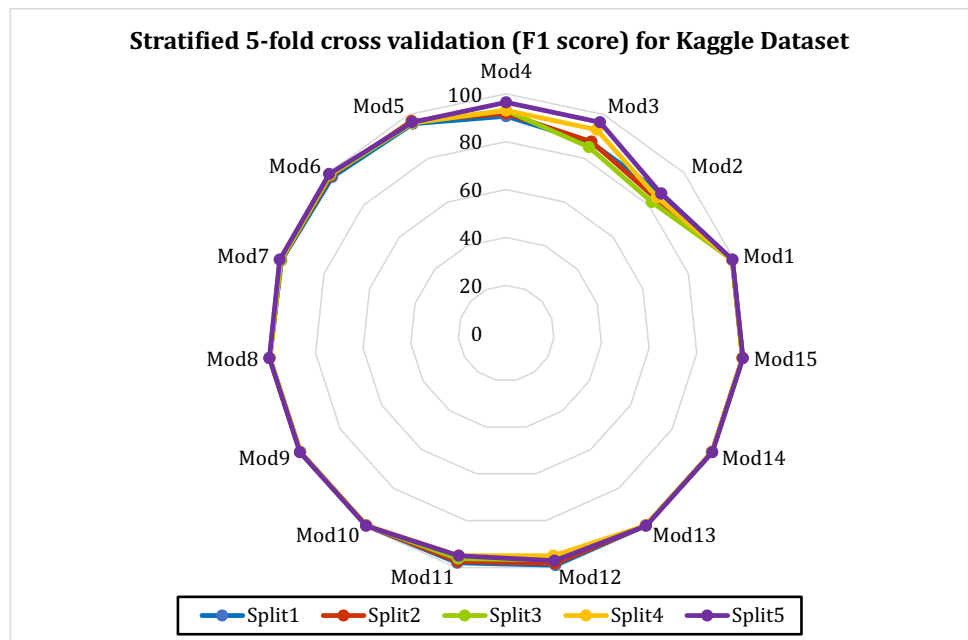


Fig. 8 Stratified 5-Fold Cross Validation (F1 score) for ‘kaggle dataset’



immediate predictions display. Figure 18 illustrates the display of the Global Health Report of a patient page.

3.4 Clinical Cases Testing Results

Table 16 summarizes the results of the clinical cases testing by reporting the rank and the presence (PRS) or not of the real diagnosis in the differential diagnosis list advanced by

our health checker using the third dataset. We noticed that both Decision Tree Model (Mod 1) and K Nearest Neighbors Model (Mod 11 to 14) show the poorest results while Random Forest Model (Mod 2 to 8), Multinomial Naïve Bayes Model (Mod 9), Logistic Regression Model (Mod 10) and Support Vector Classifier (Mod 15) show very good results in detecting the right diagnosis among the differential diagnoses list.

Fig. 9 Stratified 5-Fold Cross Validation (F1 score) for ‘first dataset’

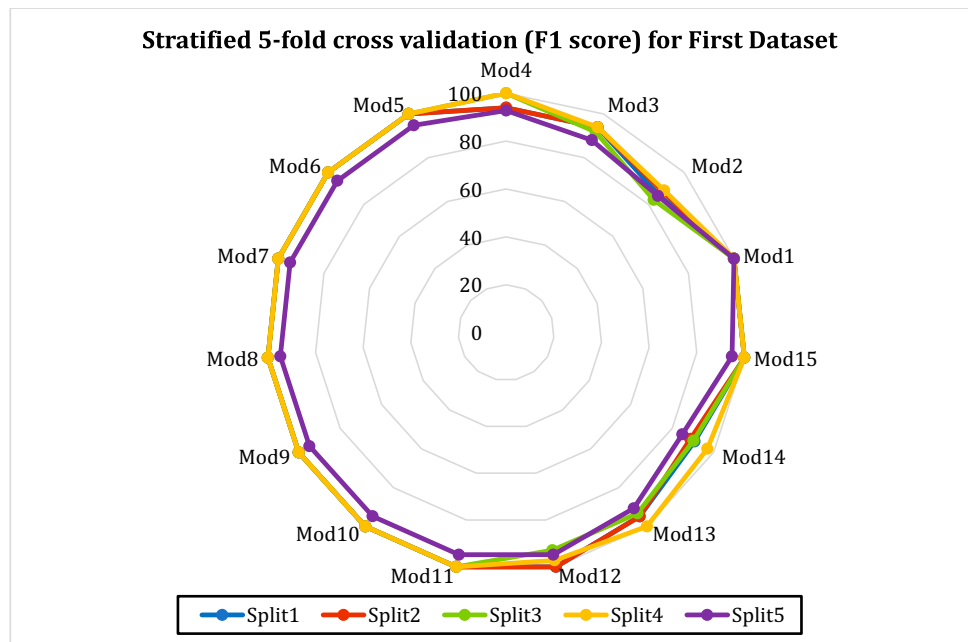
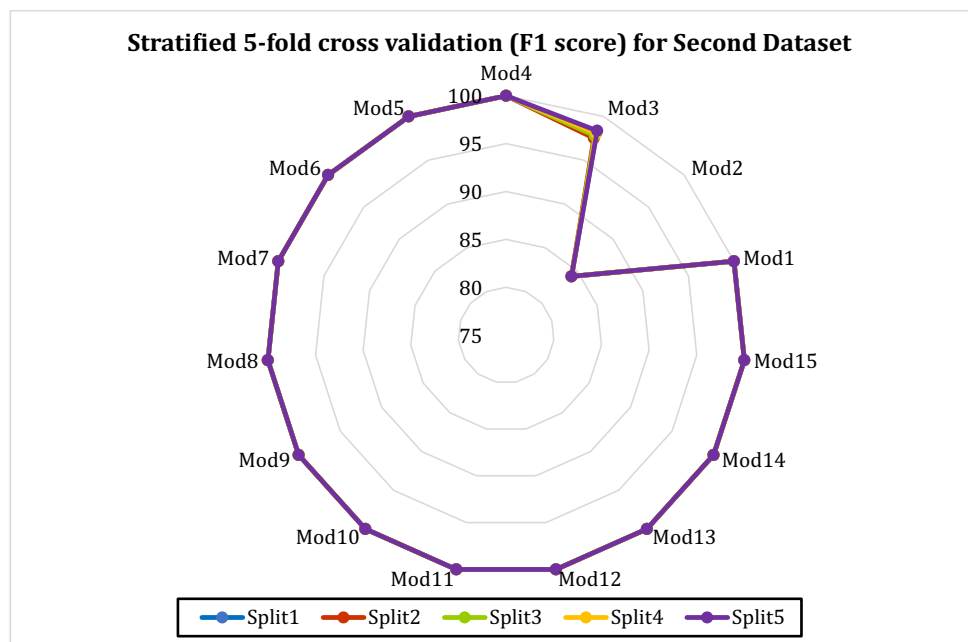


Fig. 10 Stratified 5-Fold Cross Validation (F1 score) for ‘second dataset’



4 Discussion

Expert systems for medical diagnosis or symptom checkers are kinds of software that try to imitate human doctors' clinical reasoning in order to obtain a confident and accurate symptoms-based diseases diagnosis assisting both public users looking for advice on medical issues and medical doctors to make more accurate diagnosis decisions [10, 41]. To do this, the symptom checker is meant to ask the user about his chief complaint and continue by asking about the

possible associated symptoms and other related features (medical history, age, family history allergies, etc.). Then, it generates predictions about the illnesses the user may be suffering from. Each chief complaint should be associated with a specified set of questions that need to be answered by the user in order to make the most probably accurate predictions. [20–22].

According to the literature, the most commonly used ML algorithms for diseases predictions or diagnoses are supervised algorithms including Decision Tree (DT) [43, 44], Random Forest (RF) [43, 44], Support Vector Machines

Fig. 11 Stratified 5-Fold Cross Validation (F1 score) for ‘third dataset’

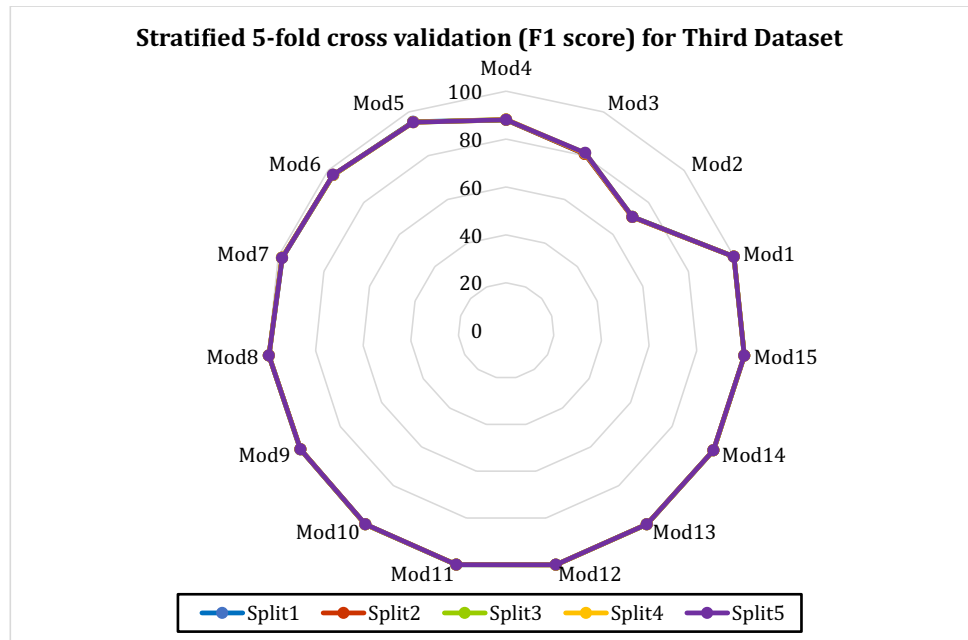
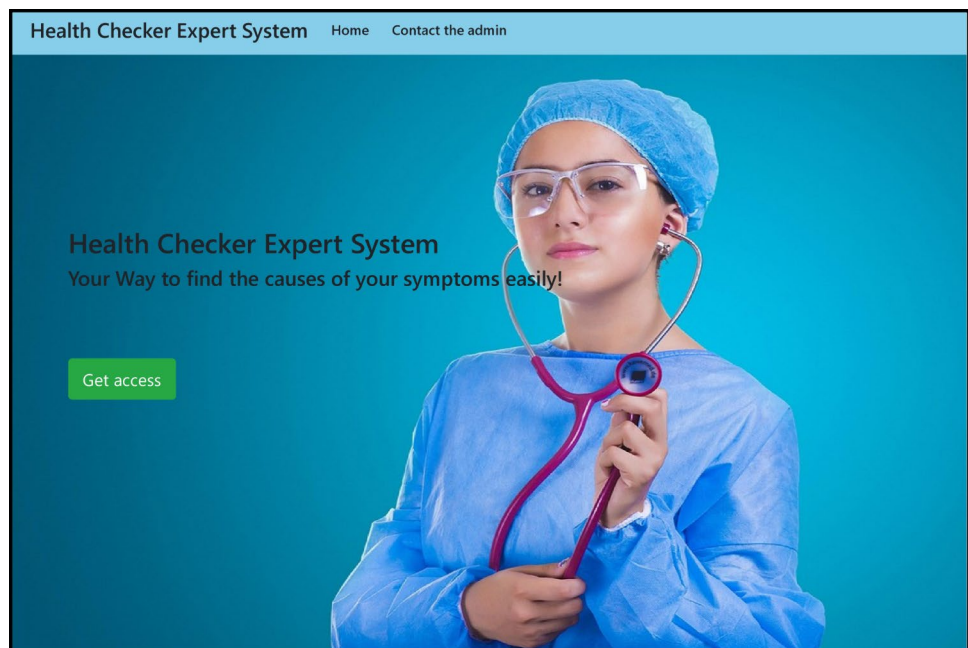


Fig. 12 Home page



(SVMs) [42], K-Nearest Neighbor (KNN) [43, 44], Naïve Bayes (NB) [42, 43], Logistic Regression (LR) [44], etc. Each one of these algorithms has its own pros and cons and selecting the optimal algorithm is fully related to the type of the problem to be solved (classification, regression...) and other factors such as: accuracy (ability of a model to correctly predict an output), speediness (time consumed in order to obtain the output prediction), training data's size (The bigger the relevant data are, the more reliable the predictions normally will be.), features' number compared with the data's size, etc. [44]

When using the 'kaggle dataset', the most of the models present an "Accuracy score" equal to 95% and an "F1 score" equal to 95% also. Practically, "Accuracy score" is enough to measure the performance of a model only in balanced datasets and it seems poorly enough to measure the performance of a model in imbalanced datasets. In the latter case, "F1 score" seems be the best performance indicator. The more the "F1 score" value tends to be "1" or "100%", the more performant is the model. In this dataset, we consider that all of the used models are performant after entering

Fig. 13 Login page

Fig. 14 Patient UI-Health Checker first page (Selecting the chief complaint step)

the optimal hyperparameters as all the models show an “F1 score” superior to 90%.

Taking into consideration the real-life medical diagnostic process and taking into account the results of the ‘kaggle dataset’ analysis, we deduced that although the ‘kaggle dataset’ shows good results from the point of view of Machine Learning, it does not serve our objective from the point of view of medical diagnosis as there are more than 42 diseases in real life, the symptoms are gathered randomly and they are not sufficiently detailed to better lead a virtual diagnosis process.

To build a fully functional symptom checker, companies generally resort to a coherent team composed of developers, data scientists and medical doctors who are in charge

of collecting medical data from clinical cases published in renowned journals [45, 46]. Their approach is scientifically well founded but it requires so much hard work and research as many difficulties may meet them when collecting the required data to obtain coherent medical datasets like the fact that the description of the patients may differ from an article to another and the team may be, thus, faced with the lack of some necessary data. Otherwise, faced with the lack of publically available trustworthy datasets that can serve the objective of building an “automatic medical diagnosis through patients’ symptoms”, some researchers have used some already existing expert systems to generate their own dataset of virtual patients. Their goal is to obtain a dataset with which they can train and test some machine learning

Fig. 15 Patient UI-Health Checker second page (Cough chief complaint form) (1)

Health Checker Expert System Home Contact the admin

Your chief complaint is: Cough

Please, remember that our symptom checker can be used only for informational purposes and it never replaces an officially medical appointment.

To find the causes of your symptoms, please select the items that apply to your case. Then, press Continue.

Gender:

- ☐ Female
- ☒ Male

Age:

- ☐ Age under 10 years
- ☐ Age between 10 and 20 years
- ☐ Age between 20 and 30 years
- ☒ Age between 30 and 40 years
- ☐ Age between 40 and 50 years
- ☐ Age between 50 and 60 years
- ☐ Age above 70 years

General information:

- ☒ Smoking
- ☐ Alcohol
- ☐ Family history
- ☐ Medical history
- ☐ Weakened immune system
- ☐ Currently taking medications
- ☐ Recent hospitalisation
- ☐ Recent travel to a risk country
- ☐ Seasonal allergy
- ☐ Medication allergy
- ☐ Exposure to toxic fumes including pollens

Fig. 16 Patient UI-Health Checker second page (Cough chief complaint form) (2)

Characteristics of the cough:

- ☐ Cough with no sputum production
- ☐ Cough with clear sputum production
- ☐ Cough with foamy sputum production
- ☐ Cough with bloody sputum production
- ☒ Cough with mucoid sputum production
- ☐ Cough with clear to purulent sputum production
- ☐ Cough with purulent sputum production

Duration of the cough:

- ☒ Cough duration less than 3 weeks
- ☐ Cough duration between 3 and 8 weeks
- ☐ Cough duration more than 8 weeks

The cough is accompanied by:

- ☐ High-grade fever
- ☐ Mild-grade fever
- ☐ Hypothermia
- ☒ Pain, tenderness, swelling and pressure around the eyes, cheeks, nose or forehead that worsens when bending over

models and so, their methodology can be efficient to solve some technical research issues and the quality of the obtained data depends on the quality of the outcomes of the used expert systems to generate data [47, 48]. In our case, researches were conducted by the developer himself who has also a medical background. Our methodology to build our dataset aims to construct a potentially trustworthy dataset that can help building an AMDS that can imitate to the maximum the real-life

medical diagnosis process. So, we have used renowned medical sources, identified a certain number of chief complaints and for each chief complaint, we selected the different possible diagnoses depending on the associated symptoms and risk factors. We judge that our approach can be a solution for the lack of publically available medical datasets if the obtained symptom checker, after being clinically tested on a representative sample of real patients, gives corrected differential

Fig. 17 Patient UI-Health Checker immediate predictions display

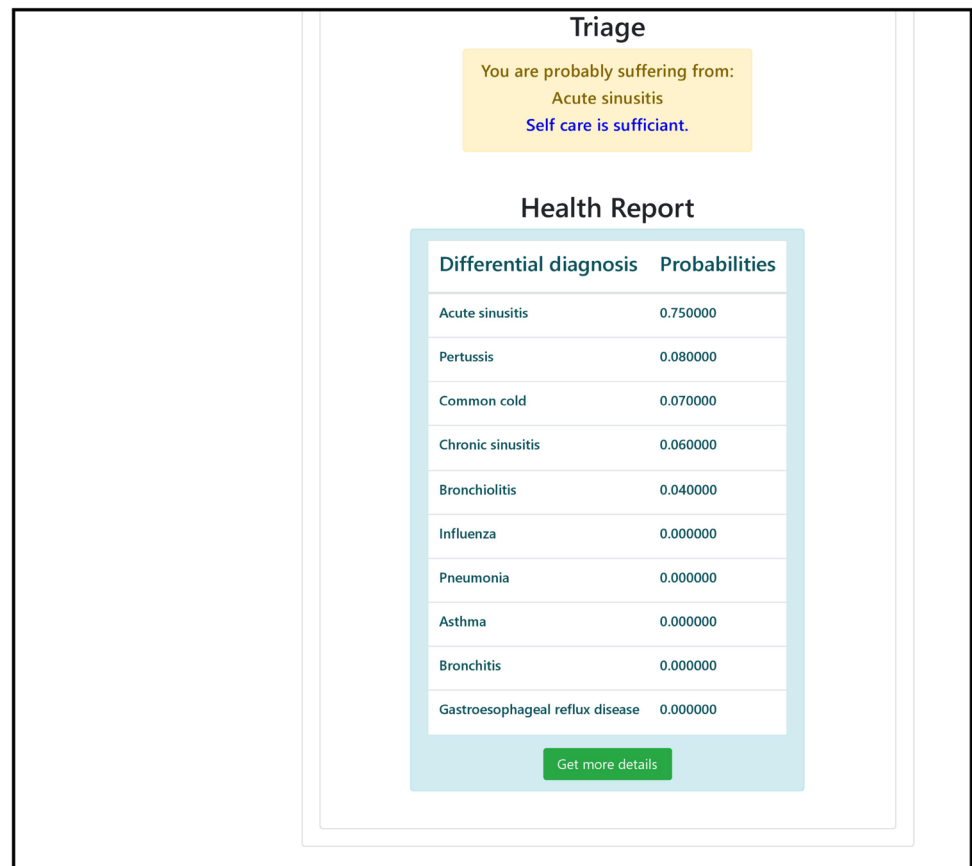
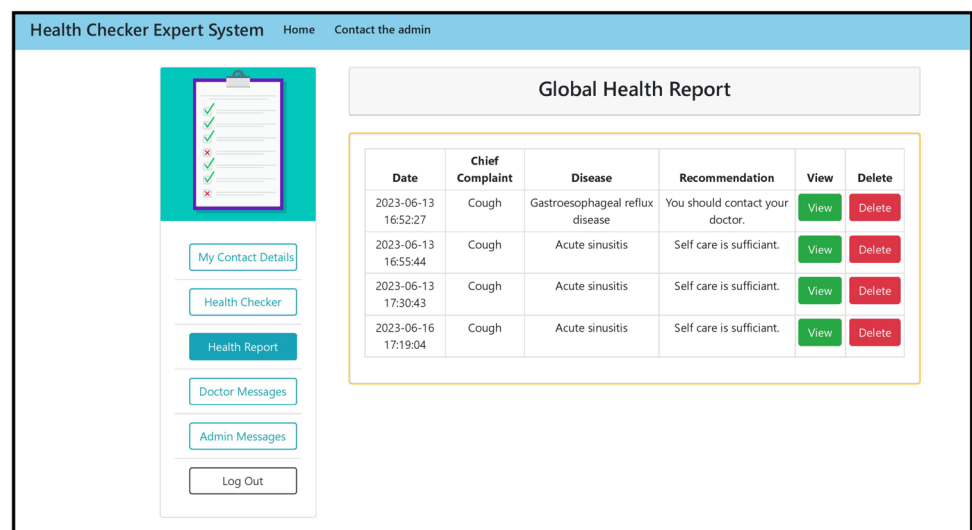


Fig. 18 Patient UI-Visualizing the Global Health Report page



diagnoses. This is preclinically proved by the results of the reported clinical cases testing (Table 16) which are in favor of the possible potential applicability of our health checker using such third dataset in real life and in consequence the validity of our approach as a solution for the lack of publicly available medical datasets leading to the construction of symptom-based effective health checkers.

In both ‘first dataset’ and ‘second dataset’, all the models may be performant as they may show an “Accuracy score” equal to 100% as well as an “F1 score” equal to 100% after adjusting the required hyperparameters. For the ‘second dataset’, we noticed also that both “Accuracy score” and “F1 score” seem be ameliorated compared with the ‘first dataset’ and this is may be because of augmenting the size of the

dataset as a result of adding more features (risk factors) and consequently adding others synthetic patients.

Upon scrutinizing the accuracy performance results of ML models on datasets in Fig. 2 and Fig. 3, several noteworthy conclusions emerge. While Kaggle and ‘first dataset’ have proven valuable in our endeavor, certain limitations must be acknowledged. Firstly, the ‘kaggle dataset’, though rich, may exhibit variability in data quality, originating from diverse sources and potentially containing inconsistencies. This heterogeneity can impact model performance, especially in real-world scenarios where data reliability is crucial. The statistics, with an accuracy ranging from 85,36% to 95,12% for most of the models and F1_scores ranging from 83,54% to 95,12% for most of the models, highlight the potential variability and underscore the need for careful consideration in leveraging this dataset for robust model development.

Additionally, ‘first dataset’, while comprehensive, may suffer from inherent biases or lack certain dimensions essential for capturing the complexity of medical diagnoses. The statistics for ‘first dataset’, with an accuracy ranging from 78,69% to 100% and F1_scores ranging from 72.71% to 100%, reveal a degree of variability in model performance. These variations suggest potential challenges in ensuring consistent and reliable predictions, urging researchers to approach the dataset with caution. These limitations underscore the need for cautious interpretation and emphasize the importance of further refining datasets for more accurate and reliable machine learning applications in the medical domain.

Taking into account both accuracy and F1_scores, ‘second dataset’ emerges as the clear frontrunner, demonstrating superiority over Kaggle and ‘first dataset’. The accuracy metrics for ‘second dataset’ consistently reach a perfect 100%, indicating the models’ precision in correctly predicting instances. Furthermore, the F1_scores for ‘second dataset’ are consistently perfect across all models, highlighting an exceptional balance between precision and recall.

Comparatively, while ‘kaggle dataset’ and ‘first dataset’ showcase commendable performance with varying accuracy and F1_scores, they fall short of achieving the consistent excellence observed in ‘second dataset’. The variability in scores for ‘kaggle dataset’ and ‘first dataset’ suggests potential challenges in maintaining a uniform level of precision and recall across different diagnoses.

The outstanding performance of ‘second dataset’, reflected in both accuracy and F1_scores, positions it as the optimal choice for robust and reliable machine learning models in the context of medical diagnosis through patients’ symptoms. The perfect F1_scores in ‘second dataset’ affirm its exceptional ability to generalize and make accurate predictions across diverse medical scenarios, reinforcing its superiority over ‘kaggle dataset’ and ‘first dataset’.

‘Second dataset’, characterized by its flawless accuracy and F1_score, stands out as a gold standard for model

generalization. The uniformity in performance across ‘second dataset’ underscores the reliability and robustness of the models in this particular dataset, highlighting their potential in real-world scenarios.

We were also careful about the risk of overfitting and model generalization. To address this, we took an additional steps by creating and evaluating larger ‘third dataset’ and performing fivefold cross-validation. This dataset was carefully designed to include a diverse range of conditions known to trigger coughing. It consisted of a substantial amount of data, with 415,712 rows representing synthetic patients. Each patient had 75 columns of features and 1 column indicating the target variable.

Overall, we evaluated several machine learning models on four different datasets: ‘kaggle dataset’, ‘first dataset’, a larger ‘second dataset’, and an even larger ‘third dataset’. The models were trained and tested using various metrics, including accuracy and F1 score.

DecisionTreeClassifier consistently achieved perfect accuracy (100%) and F1 score across all constructed datasets. This indicates that the decision tree model theoretically performed exceptionally well in classifying instances which must be confirmed or not by clinical testing as such classifier is extremely prone to overfitting.

RandomForestClassifier showed varying performance based on the dataset size and depth. For dataset1, the model with a maximum depth of 5 or more achieved perfect scores. However, on the larger ‘second dataset’, the performance decreased slightly, especially for the model with `max_depth=2`. This suggests that deeper trees may overfit on smaller datasets and generalize better on larger ones.

MultinomialNB and LogisticRegression consistently achieved high accuracy and F1 scores (both close to 100%). These models are robust and suitable for a wide range of tasks.

KNeighborsClassifier performed exceptionally well, with perfect scores across different numbers of neighbors (2, 3, 5, and 10). This demonstrates theoretically the effectiveness of the k-nearest-neighbors approach.

SVC (Support Vector Classifier) also achieved perfect scores, indicating its ability to handle complex decision boundaries. Moreover, the results were consistent and reliable through stratified fivefold cross-validation. In general, the models performed fairly well with high accuracy. The RandomForestClassifier did better than most models, especially when using deeper trees handling more complex data. Although most models performed very well in ‘second dataset’ and ‘third dataset’, models with lower tree depths in RandomForestClassifiers exhibited slightly poorer performance, indicating potential challenges in capturing and generalizing data complexities. This is proved by the results of the clinical cases testing showing better results with deeper trees.

In summary, the fivefold cross-validation was an effective way to theoretically evaluate the stability of model performance

and generalization issues while improving the reliability of model performance comparison and conducting a rigorous clinical trial on a representative sample of real patients to ensure the safety and effectiveness of our symptom checker [49] is necessary and this can be the subject of a further research. Moreover, our ongoing work [50] involves extensive testing of our health checker on real clinical cases (clinical vignettes) and benchmarking it against commercial health checkers. The results are excellent demonstrating the robustness of our study.

5 Conclusion

In conclusion, our exploration into the realm of medical diagnosis through patients' symptoms has unveiled both challenges and opportunities in leveraging machine learning for accurate and efficient predictions. We evaluated various machine learning models across four datasets using accuracy and F1 score metrics to develop a reliable health checker. The Decision-TreeClassifier consistently achieved perfect scores, while the RandomForestClassifier's performance varied based on dataset size and tree depth. Models like MultinomialNB and LogisticRegression consistently performed well, and the KNeighborsClassifier showed effectiveness across different neighbor numbers. The Support Vector Classifier also achieved perfect scores. Overall, fivefold cross-validation ensured reliable results, highlighting the RandomForestClassifier's superior performance with deeper trees. However, models with lower tree depths faced challenges in capturing data complexities. This evaluation underscores the importance of cross-validation in assessing model stability and generalization for the development of a dependable health checker. Looking ahead, our next step will be the test of our health checker with various clinical vignettes. This real-world application will provide valuable insights into the tool's practical effectiveness and usability in medical diagnosis. By evaluating the health checker's performance in clinical scenarios, we aim to validate its reliability and accuracy in a clinical setting. This process will not only enhance our understanding of its capabilities but also facilitate its integration into healthcare practices, ultimately benefiting both patients and healthcare professionals.

Authors' Contributions Leila Aissaoui Ferhi: study idea, manuscript writing and revision, dataset enhancement, choice of machine learning models, development of health checker. Manel Ben Amar: manuscript writing and revision, dataset construction, development of health checker. Fethi Choubani: manuscript revision. Ridha Bouallegue: manuscript revision.

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Data Availability On request data will be available.

Code Availability N/A.

Declarations

Ethics Approval N/A.

Consent to Participate N/A.

Consent for Publication N/A.

Conflicts of Interest/Competing Interests The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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